

WiP: Drakon: An Open Platform with Native Cellular Connectivity for XR and AI Smart Glasses

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Abstract

XR and AI smart glasses are the next mobile platform, yet most commercial devices still rely on smartphones for Internet connectivity. In this paper, we present Drakon, an open smart glass prototype with native LTE-M cellular connectivity. Drakon integrates an embedded compute platform, an LTE-M modem, onboard audio and vision sensors, and cloud services to enable multimodal AI applications. We describe the design and implementation of the system and present preliminary end-to-end AI voice interaction latency and power measurements, highlighting the performance–energy trade-offs and feasibility of standalone LTE-M-connected glasses.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; Mixed / augmented reality; • **Networks** → Network performance analysis; Network measurement; • **Hardware** → Platform power issues.

Keywords

XR and AI smart glasses, LTE-M, cellular connectivity, wearable computing, multimodal AI, mobile systems

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1 Introduction and Motivation

XR and AI smart glasses are poised to become the next mobile platform. Rather than serving merely as display devices, the next generation of smart glasses is expected to become persistent AI assistants capable of understanding the surrounding environment, answering questions about what users see, translating conversations, recognizing objects, navigating unfamiliar spaces, and providing context-aware assistance throughout the day [2]. Achieving this vision requires continuous connectivity between the glasses and cloud or edge AI services, since executing large multimodal foundation models entirely on-device remains beyond the capabilities of lightweight wearable hardware.

Status quo and limitations. Despite enormous progress, virtually all commercially available smart glasses today rely on smartphone tethering or compute puck for Internet connectivity through Bluetooth or Wi-Fi (see Figure 1). Native cellular connectivity remains largely absent from current smart glasses, fundamentally limiting their ability to operate as truly standalone computing platforms. Users must carry an additional device, remain within Bluetooth range, tolerate additional latency introduced by multi-hop communication, and accept degraded reliability whenever the smartphone is unavailable or disconnected. As AI assistants become increasingly integrated into everyday activities, this dependence on smartphones becomes a significant architectural limitation. Applications such as outdoor navigation, industrial inspection, emergency response, remote collaboration, and field operations require ubiquitous, low-latency connectivity that should be provided directly by the glass itself.

Unlike smartphones, smart glasses have extremely limited volume for batteries, antennas, and RF front-end components, while simultaneously imposing strict weight and thermal requirements for user comfort. Cellular communication is considerably more power-intensive than Bluetooth or Wi-Fi, and continuous transmission can rapidly deplete the small batteries available in wearable devices [6, 7]. Furthermore, integrating LTE or 5G antennas into the narrow temples of

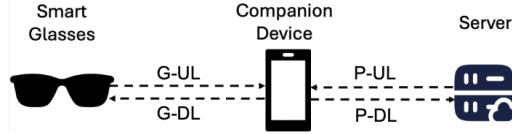


Figure 1: Communication pipeline of today's smart glasses: short-range glasses-to-phone (G-UL) and phone-to-glasses (G-DL) links connect to a companion phone, which communicates with cloud servers through phone-to-server (P-UL) and server-to-phone (P-DL) links. Most glasses rely on this phone for Internet connectivity.

eyeglasses is significantly more difficult than in smartphones due to space constraints, body-induced signal attenuation, and antenna isolation challenges [3]. These limitations have made smartphone tethering the dominant architecture for current commercial devices.

Opportunity. Fortunately, recent advances in low-power cellular technologies provide an opportunity to revisit this design space. In particular, LTE Category M1 (LTE-M) and NB-IoT [5, 7] offer substantially lower power consumption than traditional LTE while providing reasonable bandwidth for many AI-driven interactions, including voice assistants, image-based queries, notifications, sensor telemetry, and lightweight multimodal communication [1, 6, 7]. With peak data rates ranging from tens of kbps (NB-IoT) to around 1 Mbps (LTE-M), these technologies occupy a practical middle ground between ultra-low-power IoT links and broadband cellular connectivity.

Motivated by this, we present Drakon, an open experimental platform for standalone glasses with native LTE-M connectivity. It supports persistent, smartphone-free cloud access and AI interactions, including voice queries, image understanding, and multimodal assistance, within wearable power and form-factor constraints. This work-in-progress paper describes its design and implementation. The prototype is open-sourced at <https://github.com/SINRG-Lab/AIGlasses>.

2 Drakon: Prototype and Early Results

Drakon uses the Walter platform [4], combining an ESP32-S3 with a Sequans Monarch 2 GM02SP LTE-M/NB-IoT modem over UART. The ESP32-S3 handles firmware, audio, peripherals, and protocols, while the modem provides direct LTE-M connectivity. We selected it for its compute capacity, PSRAM for audio/image buffering, and mature Arduino/ESP-IDF support. The wearable layout integrates the compute/modem, audio, camera, input, antenna, and power components across the glasses' temple and bridge regions. Figure 2 shows the early prototype.

Voice capture is performed using an InvenSense INMP441 MEMS microphone. The INMP441 is a digital I²S microphone that streams mono PCM audio directly to the ESP32-S3

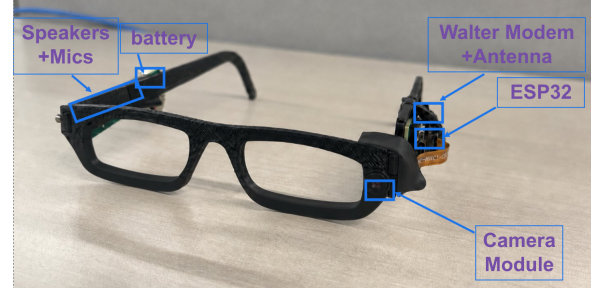


Figure 2: Drakon: A standalone LTE-M smart-glasses prototype integrating a Walter modem, ESP32-S3 controller, XIAO camera, audio subsystem, battery, antenna, and supporting electronics for untethered multimodal AI voice interaction.

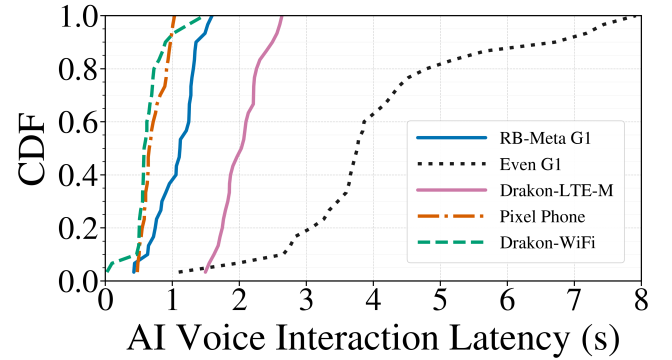


Figure 3: Comparison of AI voice interaction latency for Drakon vs. state-of-the-art platforms.

through a dedicated I²S receive interface. In our implementation, audio is sampled as 16-bit mono speech audio and buffered in PSRAM before transmission. Audio output is designed around a Maxim MAX98357A I²S Class-D amplifier connected to a miniature speaker. The firmware duplicates the mono output stream across both output channels before playback. A momentary push-button provides push-to-talk control and is sampled through an ESP32-S3 GPIO/ADC input with software debouncing and thresholding to reject idle bias and RF-coupled noise.

LTE-M communication pipeline. The ESP32-S3 communicates with the Sequans GM02SP modem through a serial AT-command interface using the WalterModem driver. During startup, the firmware initializes the modem, selects LTE-M as the radio access technology, configures the PDP context and APN settings, and waits for cellular network registration. The prototype uses a Soracom IoT SIM that provides LTE-M connectivity by attaching to a supported cellular operator (AT&T in our deployment), enabling native Internet access without smartphone tethering.

A key finding is that throughput is limited by both the cellular link and host-to-modem transfer. Because the modem's

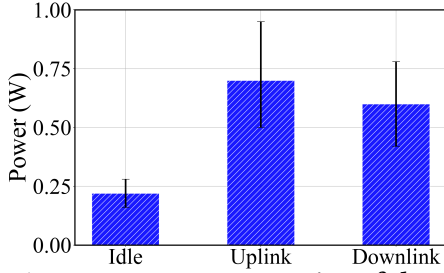


Figure 4: Average power consumption of the prototype during idle, uplink, and downlink operation.

receive ring supports reads of up to 1500 bytes, reading on every data-available event adds substantial overhead. We instead accumulate incoming data and read it in chunks approaching the 1500-byte limit.

2.1 Demonstration with AI Voice Assistant

We implemented an AI voice assistant using the OpenAI Realtime API that captures speech, streams it to the cloud, and plays synthesized responses through the speaker. We evaluate identical hardware, firmware, and AI pipelines over native LTE-M (Drakon-LTE-M) and Wi-Fi (Drakon-WiFi). First, we measure end-to-end latency from push-to-talk release to playback of the first synthesized audio sample and compare both configurations with RB-Meta G1 and Even Realities G1, which use Bluetooth-to-phone-to-Internet architectures. Second, we measure Drakon’s power consumption during idle listening, uplink transmission, and downlink reception to quantify the energy overhead of native cellular connectivity.

2.2 Latency and Power Consumption

We present preliminary measurements of end-to-end AI voice interaction latency and device-side power consumption. Latency is measured from push-to-talk release until the first synthesized audio sample is played on the glass. We compare five configurations: Drakon-LTE-M, Drakon-WiFi, a Pixel 9 phone, RB-Meta G1, and Even G1 glass.

Latency. Figure 3 compares the cumulative distribution of end-to-end AI voice interaction latency. Drakon-WiFi achieves a median latency below 1 s, closely matching a Pixel phone running the same GPT-2.1 Realtime API over both Wi-Fi and LTE-M, indicating that our wearable platform introduces little overhead beyond a conventional mobile device.

Switching from Wi-Fi to native LTE-M increases the median latency to approximately 2 s while maintaining a narrow latency distribution. Since Drakon-WiFi and Drakon-LTE-M share identical hardware, firmware, and AI pipeline, this comparison isolates the impact of LTE-M connectivity, with the additional latency primarily reflecting the current communication pipeline. The observed latency remains practical for push-to-talk AI interactions.

We also include RB-Meta G1 and Even G1 as reference points. Unlike Drakon-WiFi and Drakon-LTE-M, which both

use the same OpenAI Realtime API, these commercial devices rely on proprietary AI stacks with different models and cloud infrastructures. As a result, their end-to-end latencies also include different AI inference and backend processing times, making them useful reference points rather than direct network comparisons.

Power. Figure 4 shows Drakon’s power consumption across operating states. It consumes approximately 0.22,W while idle in discontinuous reception (DRX), 0.7,W during uplink, and 0.6,W during downlink. Because high-power operation occurs only during brief voice interactions, average consumption mainly depends on interaction frequency and time in DRX. These preliminary results demonstrate the feasibility of always-connected LTE-M for conversational AI, with further optimization possible through improved codecs, modem interfaces, and protocols.

3 Discussion and Conclusion

Drakon demonstrates the feasibility of bringing native cellular connectivity to standalone XR and AI smart glasses. While our current prototype focuses on AI voice interactions, bandwidth-intensive applications such as continuous video understanding, telepresence, and immersive XR will require capabilities beyond what LTE-M alone can provide.

We envision future systems leveraging AI-on-RAN infrastructure, where low-resolution video or compact visual representations are transmitted from the glasses and enhanced using GPU resources before being forwarded to cloud services or remote users. Such cross-layer designs can combine glasses with intelligent network infrastructure to enable richer experiences.

We believe Drakon is an important first step toward truly standalone cellular-connected smart glasses and hope it will serve as an open platform for future research on networking and edge AI for smart glasses.

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